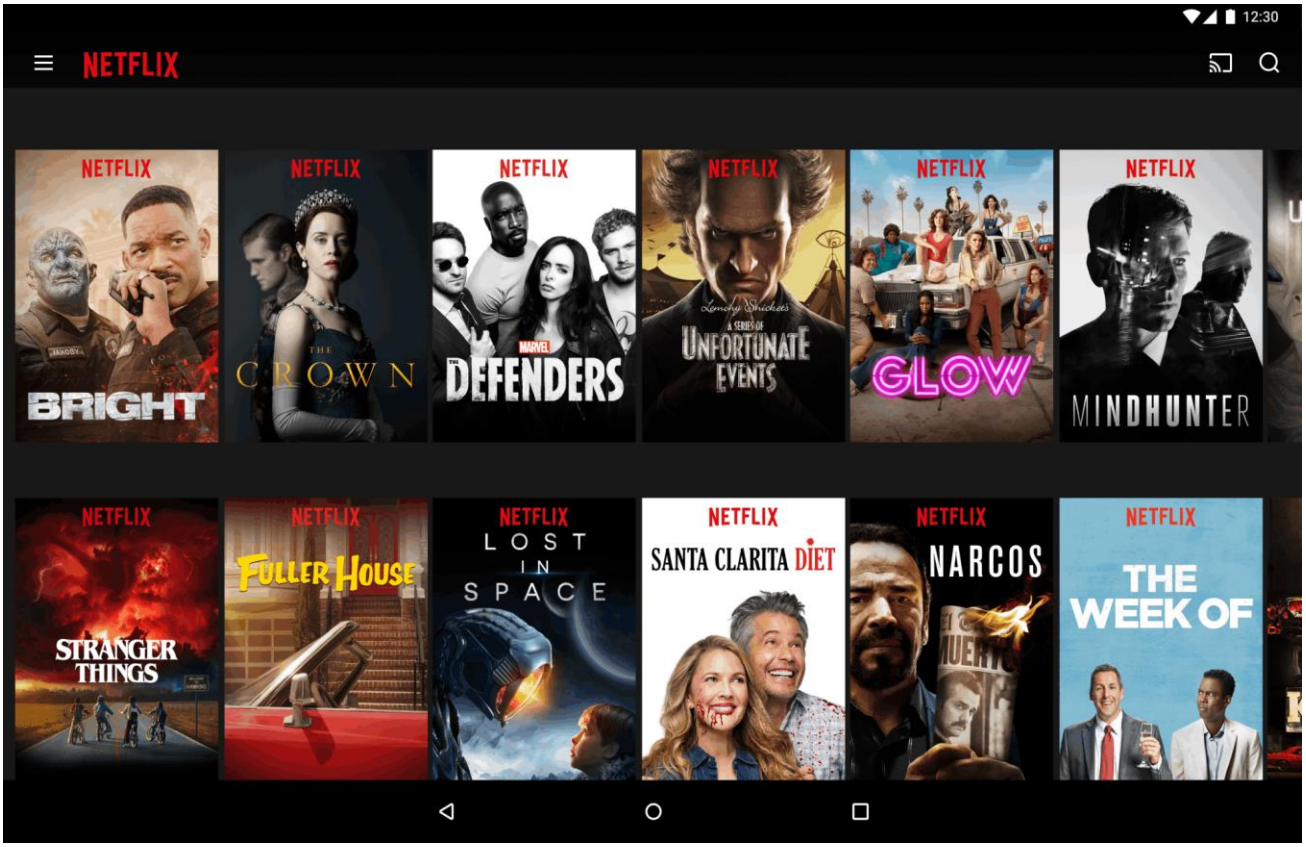


Netflix

1. Project Overview

This project presents an exploratory data analysis (EDA) of Netflix content using Python.

The main objective is to explore the characteristics of movies and TV shows, identify patterns in content distribution, and investigate trends related to release years, countries, genres, content ratings, and duration.

The analysis was conducted using a dataset containing 8,807 titles, including 6,131 movies and 2,676 TV shows.

Python, Pandas, and Matplotlib were used for data cleaning, transformation, statistical analysis, and data visualisation.

The project demonstrates practical skills in data preprocessing, exploratory analysis, data visualisation, and communicating insights.

2. Dataset Description

The dataset contains information about Netflix movies and TV shows, with 8,807 records and 12 original columns.

The dataset includes the following information:

- **show_id:** Unique identifier for each title.
- **type:** Content type (Movie or TV Show).
- **title:** Name of the movie or TV show.
- **director:** Director's name.
- **cast:** Main actors and actresses.
- **country:** Country associated with the production.
- **date_added:** Date when the title was added to Netflix.
- **release_year:** Original release year.
- **rating:** Content age classification.
- **duration:** Movie duration in minutes or TV show seasons.
- **listed_in:** Content categories and genres.
- **description:** Brief description of the title.

The dataset contains 6,131 movies (69.62%) and 2,676 TV shows (30.38%).

During data preprocessing, additional variables were created, including the year and month when titles were added to Netflix.

These variables were used to support temporal analysis and identify patterns in the dataset.

https://www.kaggle.com/datasets/swatikhedekar/exploratory-data-analysis-on-netflix-data

3. Methodology

The project followed an exploratory data analysis (EDA) approach, including data inspection, cleaning, transformation, statistical analysis, and visualisation.

3.1 Tools and Technologies

The following tools were used:

- **Python:** Data processing and analysis.

- **Pandas:** Data manipulation, cleaning, and statistical calculations.
- **Matplotlib:** Data visualisation.
- **Visual Studio Code:** Python development environment.

3.2 Data Inspection

The initial dataset contained 8,807 records and 12 columns.

The data inspection process included:

- Examining the dataset structure and data types.
- Identifying missing values and calculating their percentages.
- Checking for duplicate records, titles, and identifiers.
- Investigating inconsistent values in the rating and duration columns.

No duplicate records, titles, or show IDs were identified.

3.3 Data Cleaning

Missing values were identified in several columns, including director, cast, country, date_added, rating, and duration.

The cleaning process included handling missing categorical values and converting date_added into a datetime format.

Three movie records contained duration values incorrectly stored in the rating column. These inconsistencies were identified and corrected.

Ten records with missing addition dates were retained, with their dates remaining unavailable.

3.4 Feature Engineering

Additional variables were created to support the analysis:

- year_added: Year when the title was added to Netflix.
- month_added: Month when the title was added.
- duration_minutes: Movie duration converted into numerical values.
- seasons: Number of seasons extracted from TV show duration values.

3.5 Exploratory Data Analysis

The analysis investigated content distribution, release years, countries, genres, age classifications, movie durations, and TV show seasons.

Descriptive statistics and visualisations were used to identify patterns and summarise the dataset.

4. Exploratory Data Analysis

4.1 Content Distribution: Movies vs TV Shows

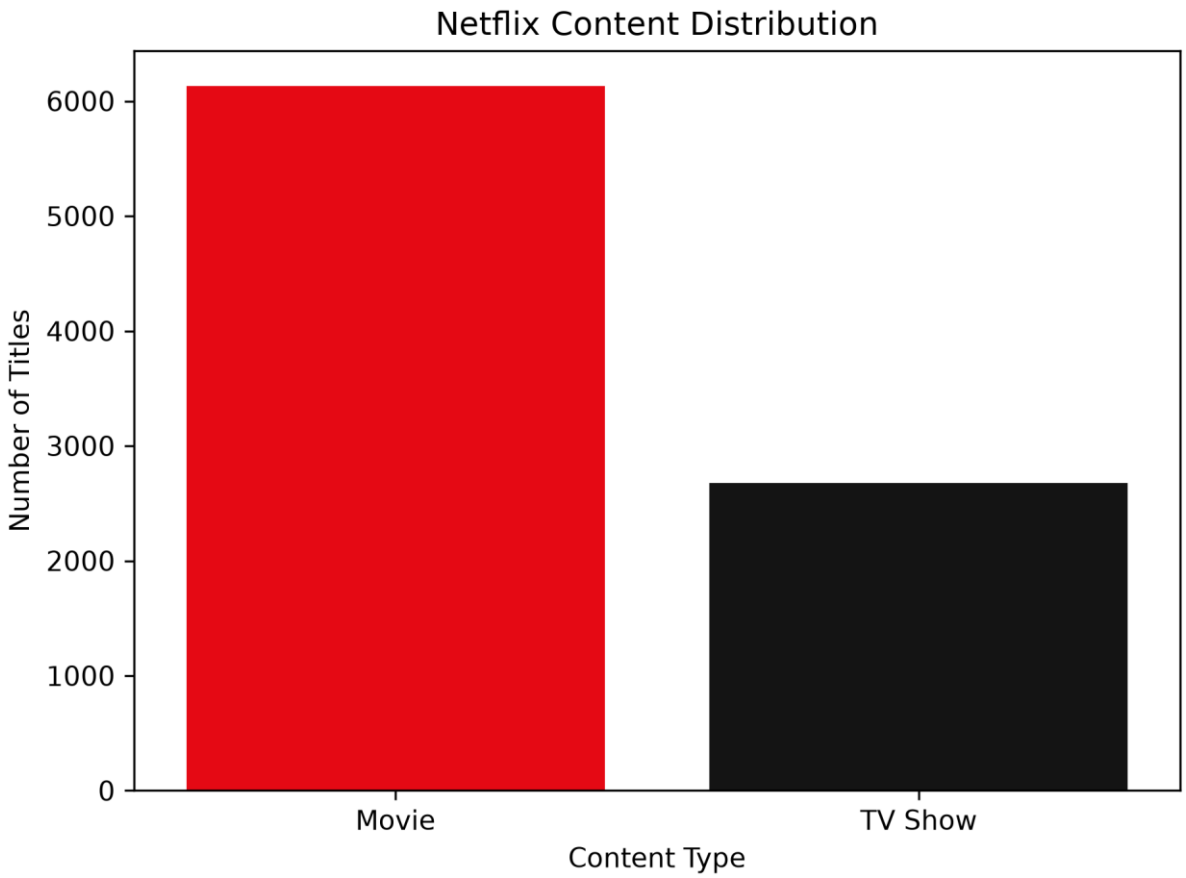
The analysis identified 8,807 titles in the dataset, consisting of 6,131 movies and 2,676 TV shows.

Movies represent 69.62% of the catalog, while TV shows account for 30.38%.

Key Insight:

Movies represent the majority of titles in the dataset, with approximately seven out of every ten titles classified as movies.

This finding demonstrates a substantial concentration of movie content within the catalog represented by the dataset.



4.2 Content by Release Year

This analysis examines the distribution of movies and TV shows according to their original release years.

Movie titles reached their highest count in 2017 and 2018, with 767 titles in each year.

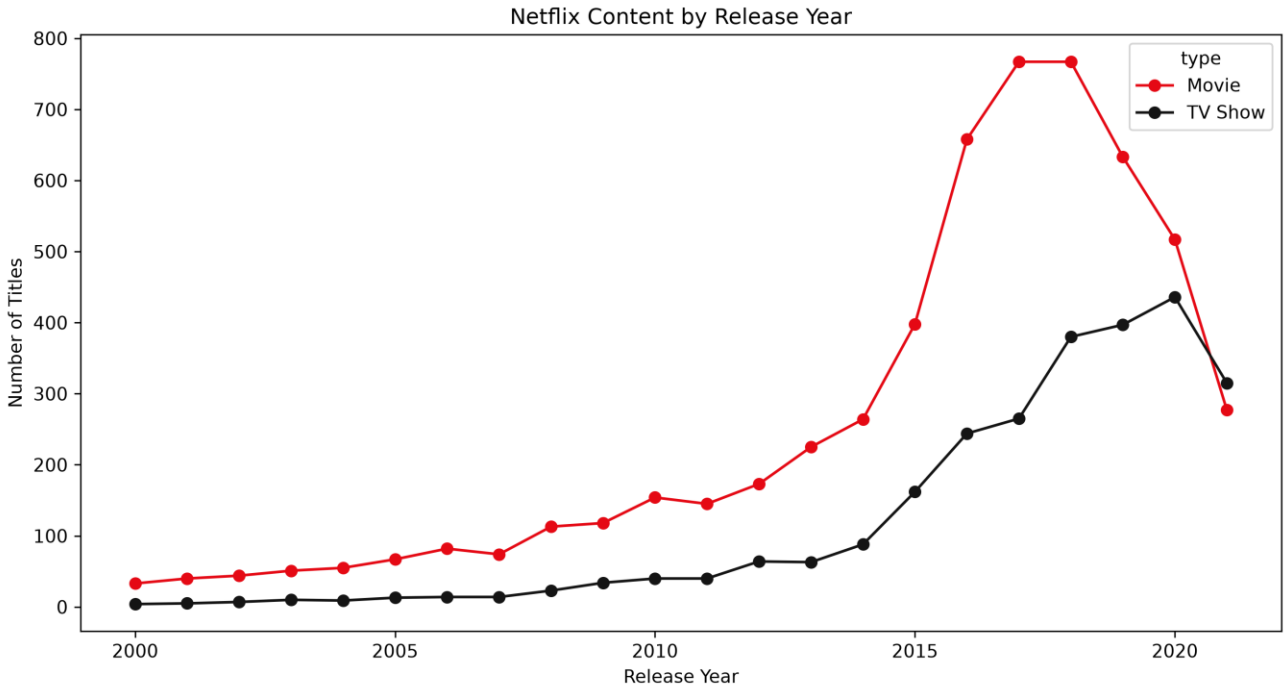
TV shows reached their highest recorded count in 2020, with 436 titles.

Key Insights

- Movies were the predominant content type across most release years.
- Movie titles peaked in 2017 and 2018.
- TV shows reached their highest count in 2020.

Limitation

The dataset may have incomplete coverage of recent releases, particularly in 2021. These figures represent titles included in the dataset, not total worldwide productions.



4.3 Content Addition Trends

This analysis examines the number of movies and TV shows added to Netflix between 2008 and 2021.

The dataset shows a substantial increase in content additions from 2016 onwards.

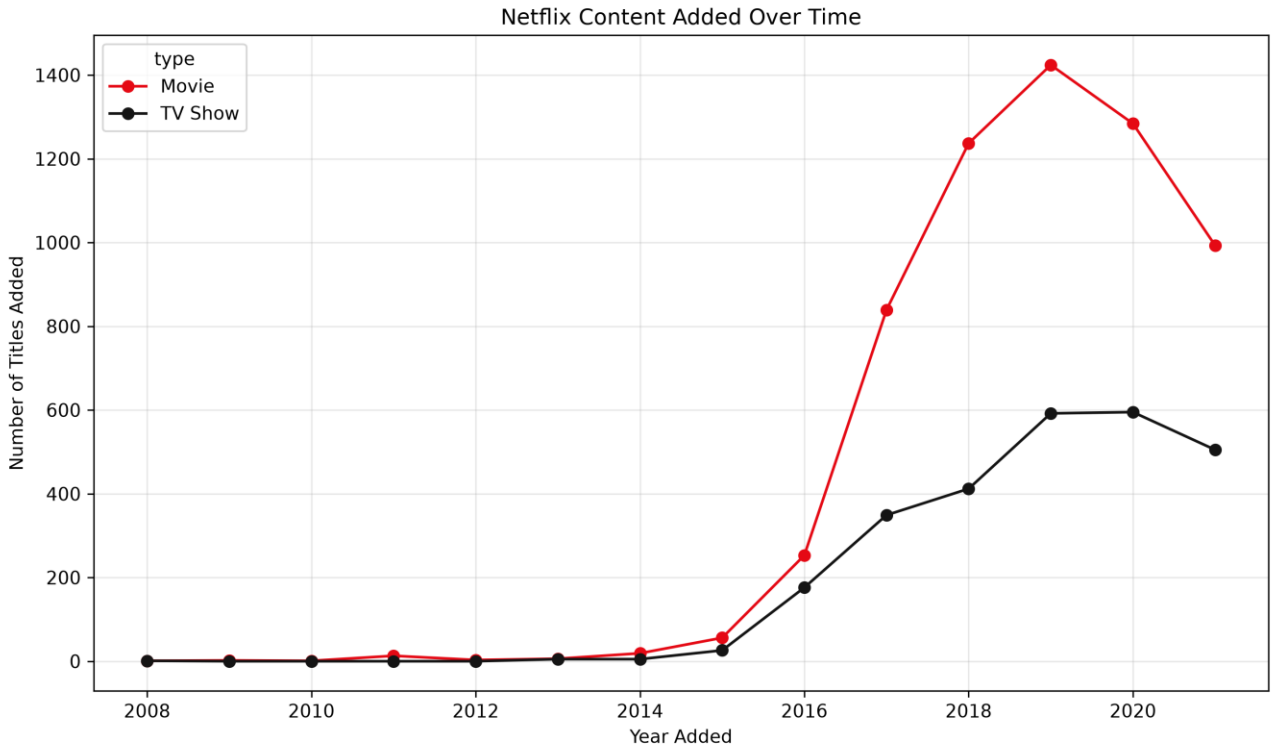
Movie additions reached their highest recorded level in 2019, with 1,424 titles. TV show additions peaked in 2020, with 595 titles.

Key Insights

- Movie additions peaked in 2019, with 1,424 titles.
- TV show additions peaked in 2020, with 595 titles.
- The dataset shows a substantial increase in content additions between 2016 and 2019.

Limitations

The dataset may not include all titles added during 2021. Therefore, the lower counts in that year should not be interpreted as evidence of a decline in Netflix's content acquisition strategy.



4.4 Top 10 Countries by Number of Titles

This analysis identifies the countries most frequently associated with titles in the Netflix dataset.

The United States leads with 3,690 titles, followed by India with 1,046 and the United Kingdom with 806.

Canada and France complete the top five, with 445 and 393 titles, respectively.

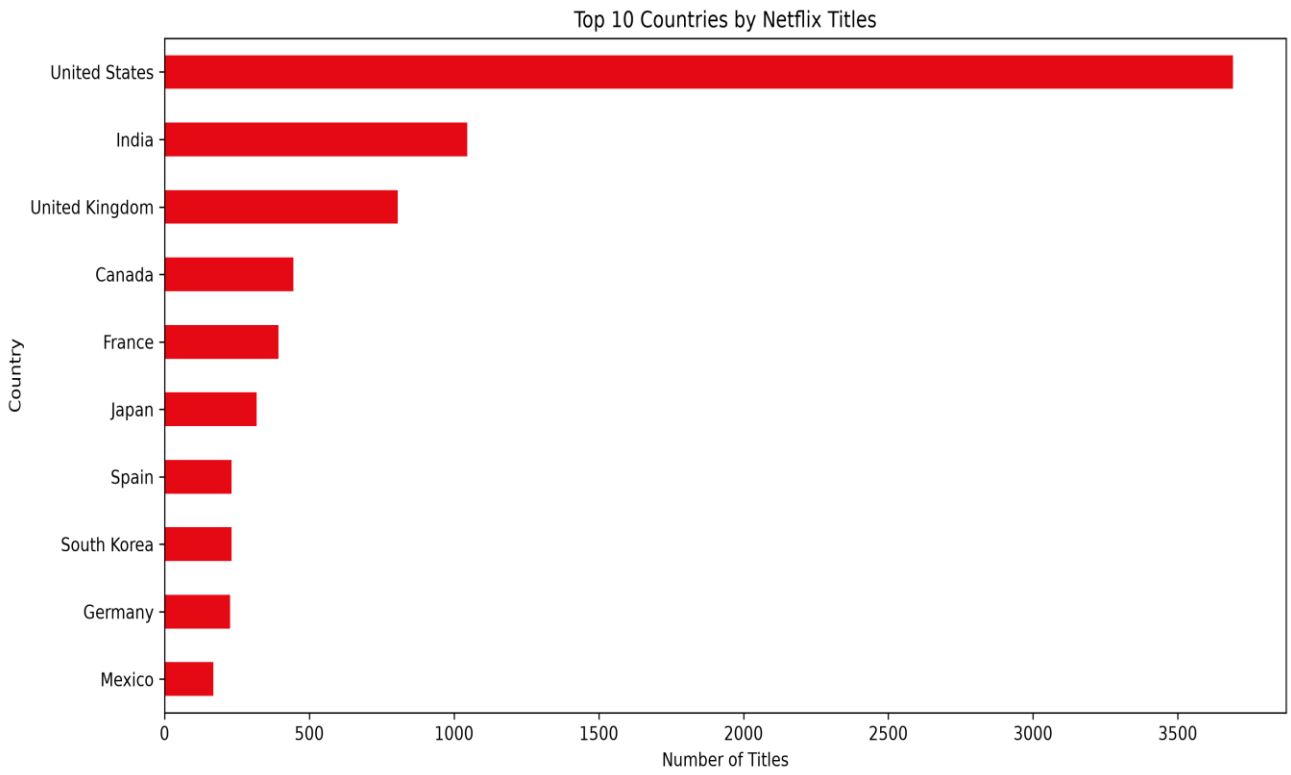
Key Insights

- The United States has the highest number of associated titles in the dataset.
- India and the United Kingdom have the second- and third-highest counts.
- The dataset includes titles associated with countries across different regions.

Methodological Note

Some titles are associated with multiple countries. Each country was counted separately when analysing production-country participation.

Therefore, country counts may overlap and should not be interpreted as mutually exclusive title totals.



4.5 Top 10 Content Categories

This analysis examines the most frequently listed content categories in the Netflix dataset.

International Movies is the most common category, appearing in 2,752 titles, followed by Dramas with 2,427 titles and Comedies with 1,674 titles.

International TV Shows ranks fourth, with 1,351 titles, followed by Documentaries with 869 titles.

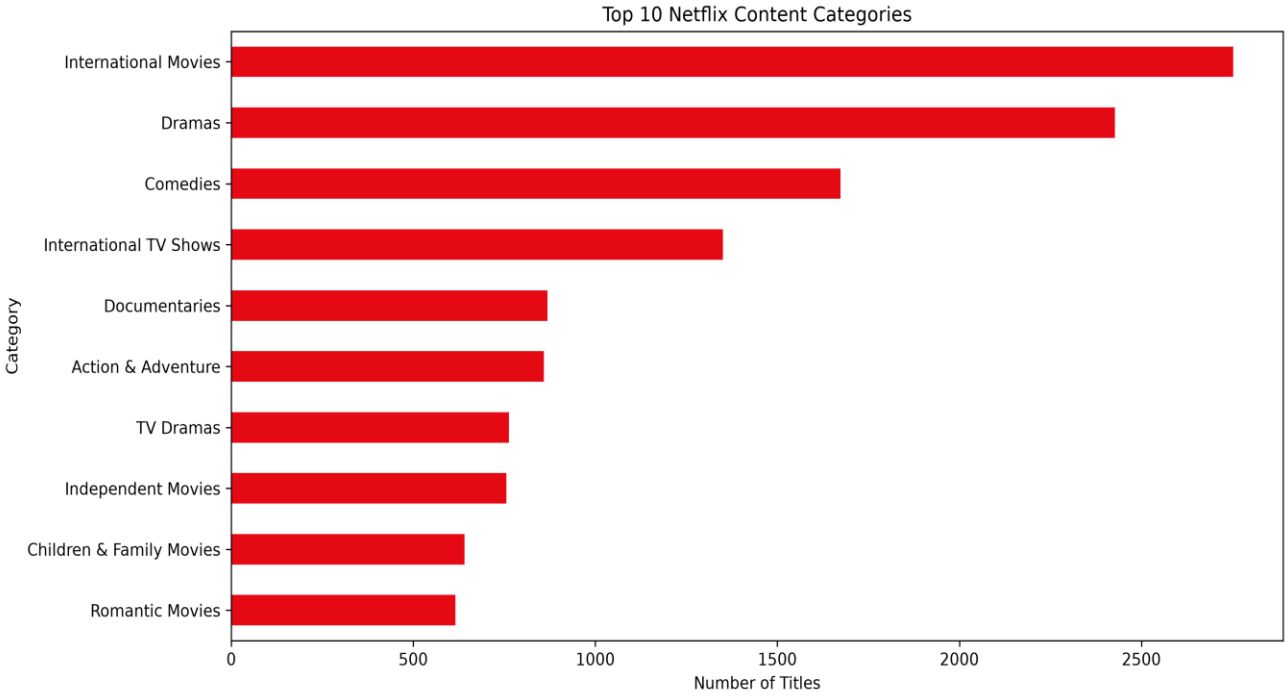
Key Insights

- International Movies is the most frequently listed category, with 2,752 titles.
- Dramas and Comedies are also prominent, appearing in 2,427 and 1,674 titles, respectively.
- International TV Shows is the most frequent TV-specific category, with 1,351 titles.
- The distribution highlights the variety of content categories represented in the dataset.

Methodological Note

The `listed_in` column contains multiple categories for some titles. These values were separated and counted individually using Python and Pandas.

As a result, a single title may contribute to more than one category. Category counts overlap and should not be summed to calculate the total number of unique titles.



4.6 Content Ratings Analysis

This analysis examines the distribution of content ratings in the Netflix dataset to identify the most frequently assigned age classifications.

TV-MA is the most common rating, appearing in 3,207 titles (36.44% of the dataset), followed by TV-14 with 2,160 titles (24.55%) and TV-PG with 863 titles (9.81%).

The R rating appears in 799 titles, while PG-13 accounts for 490 titles.

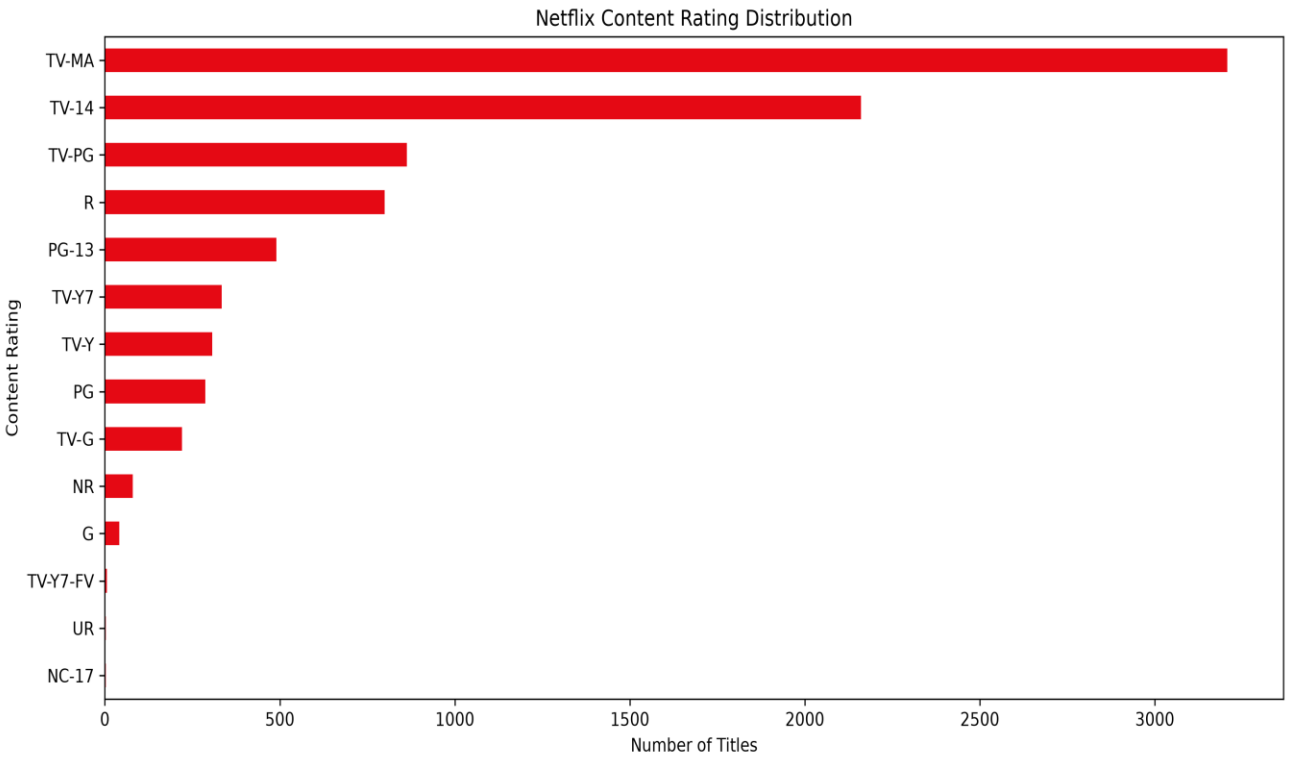
Key Insights

- TV-MA is the most frequent rating, representing 36.44% of all titles.
- TV-14 is the second most common rating, accounting for 24.55%.
- Together, TV-MA and TV-14 represent approximately 61% of the dataset.
- The distribution shows a substantial concentration of titles in these two rating categories.

Methodological Note

The analysis was conducted using the `rating` column. Missing and invalid rating values were identified during data cleaning and should be treated as unknown rather than assigned an age classification.

Ratings describe content classifications and do not directly measure audience demographics or actual viewing behaviour.



4.7 Movie Duration Analysis

This analysis examines the duration of movies in the Netflix dataset to identify typical running times and understand how movie lengths are distributed.

The dataset contains 6,131 movies, with an average duration of approximately 99.6 minutes and a median duration of 98 minutes.

The shortest movie is 3 minutes long, while the longest is 312 minutes. Most movies are concentrated around the 90–120-minute range.

Key Insights

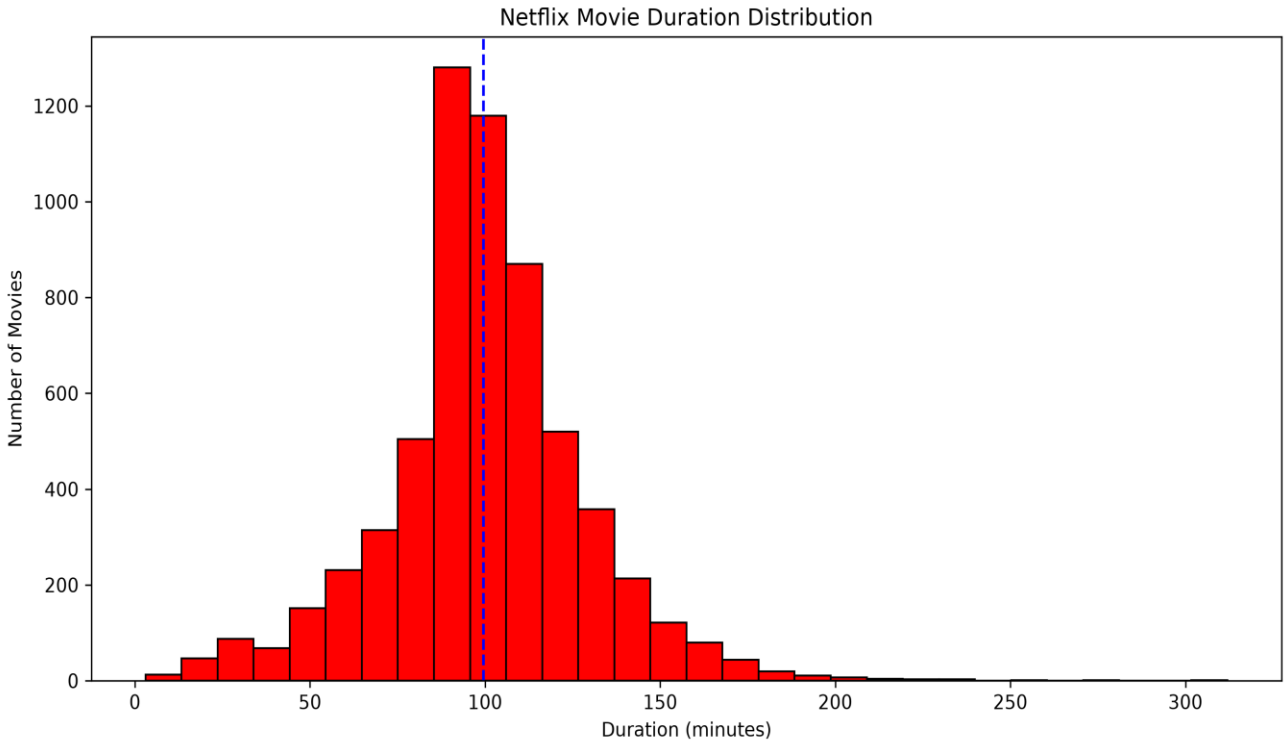
- The average movie duration is approximately 100 minutes.
- The median duration is 98 minutes, indicating that half of the movies have a duration of 98 minutes or less.
- The middle 50% of movies have durations between 87 and 114 minutes.
- The distribution has a right tail, indicating that a smaller number of particularly long movies extend the upper end of the duration range.

Methodological Note

Movie durations were extracted from the `duration` column and converted into numerical values representing minutes.

During data cleaning, three movie records were identified with duration values incorrectly stored in the `rating` column. These values were recovered and assigned to the correct duration field.

After correction, all 6,131 movies had valid duration values for this analysis.



4.8 TV Show Seasons Analysis

This analysis examines the number of seasons associated with TV shows in the Netflix dataset.

The dataset contains 2,676 TV shows. Of these, 1,793 titles have only one season, representing approximately 67% of all TV shows.

Another 425 titles have two seasons, while 199 have three seasons. The number of titles generally decreases as the season count increases.

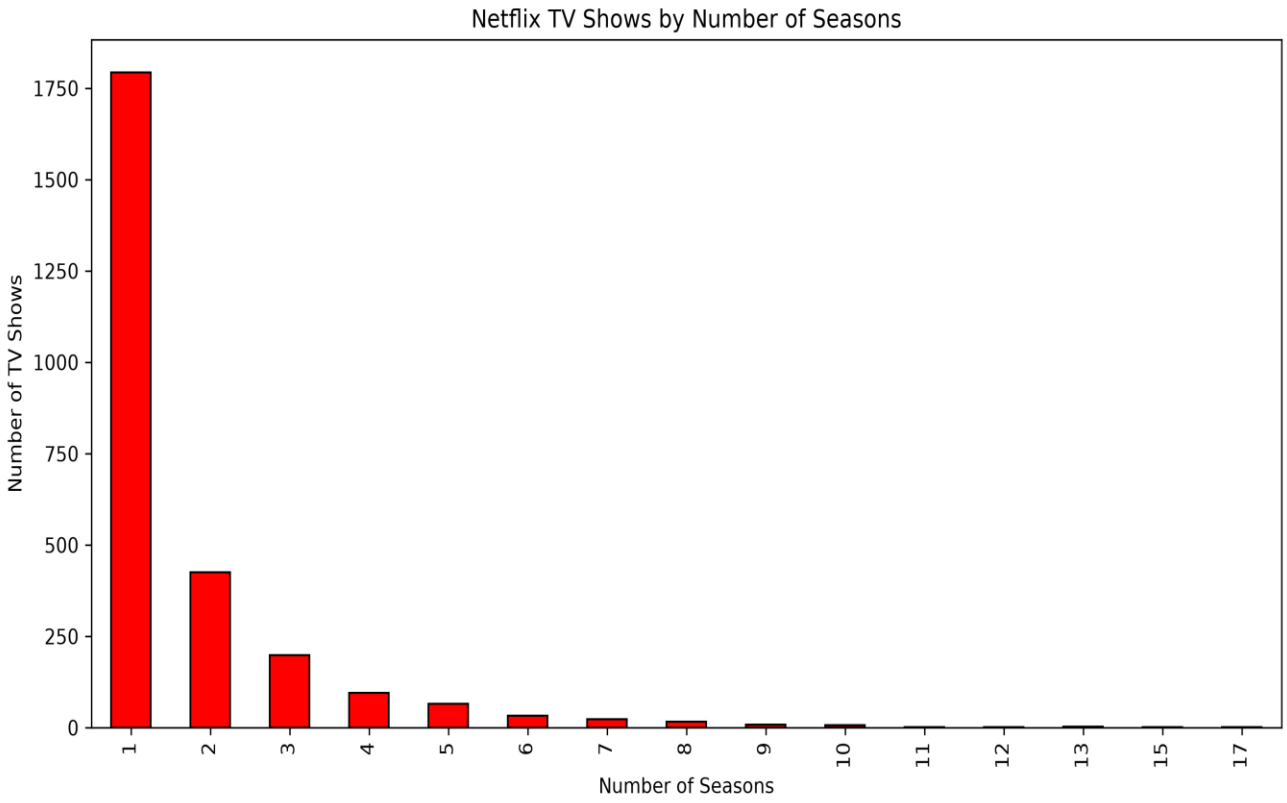
Key Insights

- One-season TV shows account for approximately 67% of all TV shows in the dataset.
- Two-season shows are the second most common group, with 425 titles.
- Shows with three or more seasons are less frequent.
- The distribution is concentrated among titles with fewer seasons.

Methodological Note

The number of seasons was extracted from the `duration` column and converted into numerical values.

Season counts describe the information recorded in the dataset. A title having only one season does not necessarily mean it was cancelled, as it may be a limited series or have incomplete catalog information.



4.9 Comparative Analysis: Movies vs TV Shows

This analysis compares movies and TV shows across the Netflix dataset to identify differences in content distribution, release patterns, and duration characteristics.

The dataset contains 6,131 movies (69.62%) and 2,676 TV shows (30.38%), indicating that movies represent the majority of recorded titles.

The release-year analysis shows that movie titles reached their highest count in 2017 and 2018, with 767 titles in each year. TV shows reached their highest count in 2020, with 436 titles.

The duration analysis also reveals different characteristics between the two content types. Movies have an average duration of approximately 100 minutes, while 67% of TV shows have only one season.

Key Insights

- Movies account for approximately 70% of the dataset.
- Movie titles peaked in the release-year analysis in 2017 and 2018.
- TV show titles reached their highest release-year count in 2020.
- Movies have an average duration of approximately 100 minutes.
- Most TV shows in the dataset have only one season.

Methodological Note

Movies and TV shows were compared using the `type` column.

Release-year counts represent titles included in the dataset, rather than total worldwide productions.

Movie durations are measured in minutes, while TV show durations are measured in seasons. These measurements were analysed separately because they represent different units.

5. Key Findings

The exploratory data analysis of 8,807 Netflix titles revealed several patterns related to content distribution, production countries, categories, age classifications, and duration.

Main Findings

- Content Distribution: Movies represent 69.62% of the dataset, while TV shows account for 30.38%.
- Content Addition Trends: Movie additions peaked in 2019, with 1,424 titles, while TV show additions reached their highest recorded level in 2020, with 595 titles.
- Geographical Distribution: The United States has the highest number of associated titles (3,690), followed by India (1,046) and the United Kingdom (806).
- Content Categories: International Movies is the most frequently listed category, appearing in 2,752 titles, followed by Dramas (2,427) and Comedies (1,674).
- Content Ratings: TV-MA is the most common age classification, representing approximately 36.44% of the dataset.
- Movie Duration: Movies have an average duration of approximately 100 minutes and a median duration of 98 minutes.
- TV Show Seasons: Approximately 67% of TV shows have only one season recorded in the dataset.

Overall Interpretation

The analysis reveals a dataset predominantly composed of movies, with substantial representation from the United States and a high frequency of international content categories.

The results also highlight differences between movies and TV shows in their temporal distribution and duration characteristics.

These findings describe the titles represented in the dataset and should not be interpreted as direct evidence of Netflix's global audience preferences, business performance, or complete catalog composition.

6. Limitations

Although the analysis provides useful insights into the Netflix dataset, several limitations should be considered when interpreting the results.

Dataset Coverage

The dataset contains 8,807 titles and may not represent Netflix's complete or current catalog. In particular, the records for 2021 may be incomplete.

Missing Data

Several columns contained missing values, including director, cast, country, and date_added. These missing values may affect analyses involving those variables.

Multiple Categories and Countries

Some titles are associated with multiple countries and content categories. Consequently, the corresponding counts overlap and cannot be interpreted as mutually exclusive totals.

Limited Business Information

The dataset does not include viewing figures, subscriber behaviour, revenue, or audience engagement metrics. Therefore, the analysis cannot determine content popularity, profitability, or viewer preferences.

Temporal Interpretation

The `release_year` column represents the original release year, while `year_added` represents when a title was added to Netflix. These variables describe different events and were analysed separately.

7. Conclusion

This project conducted an exploratory data analysis of 8,807 Netflix titles using Python, Pandas, and Matplotlib.

The analysis identified patterns in content distribution, release years, catalog additions, production countries, content categories, age classifications, movie durations, and TV show seasons.

Movies accounted for approximately 70% of the dataset, while TV shows represented approximately 30%. The United States had the highest number of associated titles, and International Movies was the most frequently listed content category.

The project also demonstrated the importance of data cleaning, particularly when handling missing values and correcting inconsistencies in the duration and rating columns.

Overall, this project strengthened practical skills in data preprocessing, exploratory data analysis, statistical interpretation, data visualisation, and communicating findings.

The cleaned dataset and analytical results can also support the development of an interactive Power BI dashboard.